SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base
1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks
1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.
1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal
Prove that **the user is granted access** using the Resolution Algorithm.
Tasks
1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet using the Resolution Algorithm.

1. Representation in Propositional Logic

Let:

- (R) = It is raining.
- (W) = The ground is wet.

Statement 1

If it rains, then the ground becomes wet.

[
R \rightarrow W
]

Statement 2

It is raining.

[
R
]

Goal

[
W
]

2. Conversion into CNF

The implication rule is:

[
P \rightarrow Q \equiv \neg P \lor Q
]

Therefore,

[
R \rightarrow W
]

becomes:

[
\neg R \lor W
]

The statement (R) is already in CNF.

Thus, the CNF clauses are:

[
C_1 = \neg R \lor W
]

[
C_2 = R
]

3. Negation of the Goal

The goal is:

[
W
]

For resolution proof by contradiction, negate the goal:

[
\neg W
]

Therefore, the clauses for resolution are:

[
C_1 = \neg R \lor W
]

[
C_2 = R
]

[
C_3 = \neg W
]

4. Resolution Algorithm – Step by Step

Resolution Step 1

Take:

[
C_1 = \neg R \lor W
]

and

[
C_2 = R
]

The complementary literals are:

[
\neg R \quad \text{and} \quad R
]

Using the Resolution Rule:

[
\frac{\neg R \lor W \qquad R}{W}
]

Therefore, the new clause is:

[
\boxed{C_4 = W}
]

Resolution Step 2

Now take:

[
C_4 = W
]

and the negated goal:

[
C_3 = \neg W
]

The complementary literals are:

[
W \quad \text{and} \quad \neg W
]

Using the Resolution Rule:

[
\frac{W \qquad \neg W}{\boxed{\square}}
]

Therefore, we obtain the Empty Clause**:**

[
\boxed{\square}
]

5. Complete Resolution Proof

The complete proof is:

[
\frac{\neg R \lor W \qquad R}{W}
]

Then,

[
\frac{W \qquad \neg W}{\boxed{\square}}
]

Thus,

[
\boxed{\square}
]

is obtained.

6. Conclusion

The Empty Clause (\boxed{\square}) indicates that the negation of the goal leads to a contradiction.

Therefore, the original goal is true.

[
\boxed{W}
]

Hence,

[
\boxed{\text{The ground is wet.}}
]

Therefore, the statement “The ground is wet” is successfully proved using the Resolution Algorithm.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks using the Resolution Algorithm**.**

1. Representation in Propositional Logic

Let:

- (S) = Rahul submitted the assignment.
- (M) = Rahul receives internal marks.

Statement 1

If a student submits the assignment, then the student receives internal marks.

[
S \rightarrow M
]

Statement 2

Rahul submitted the assignment.

[
S
]

Goal

[
M
]

2. Conversion of Premises into CNF

The implication rule is:

[
P \rightarrow Q \equiv \neg P \lor Q
]

Therefore,

[
S \rightarrow M
]

is converted into:

$$
\neg S \lor M
$$

The second premise is already in CNF:

$$
S
$$

Thus, the CNF clauses are:

$$
C_1 = \neg S \lor M
$$

$$
C_2 = S
$$

3. Negation of the Goal

The goal is:

$$
M
$$

For resolution proof by contradiction, negate the goal:

$$
\neg M
$$

Therefore, the complete set of clauses is:

$$
C_1 = \neg S \lor M
$$

[
C_2 = S
]

[
C_3 = \neg M
]

4. Apply the Resolution Algorithm

Resolution Step 1

Resolve (C_1) and (C_2):

[
C_1 = \neg S \lor M
]

[
C_2 = S
]

The complementary literals are:

[
\neg S \quad \text{and} \quad S
]

Applying the Resolution Rule:

[
\frac{\neg S \lor M \qquad S}{M}
]

Therefore, the new clause is:

[
\boxed{C_4 = M}
]

Resolution Step 2

Now resolve (C_4) with the negated goal (C_3):

$$
C_4 = M
$$

$$
C_3 = \neg M
$$

The complementary literals are:

$$
M \quad \text{and} \quad \neg M
$$

Applying the Resolution Rule:

$$
\frac{M \qquad \neg M}{\boxed{\square}}
$$

Therefore, we obtain the Empty Clause**:**

$$
\boxed{\square}
$$

5. Complete Resolution Proof

The complete resolution process is:

$$
\frac{\neg S \lor M \qquad S}{M}
$$

Then:

$$
\frac{M \qquad \neg M}{\boxed{\square}}
$$

Hence, the Empty Clause is obtained:

[
\boxed{\square}
]

6. Conclusion

Since the Empty Clause (\square) is obtained, the negation of the goal leads to a contradiction.

Therefore, the original goal is proved.

[
\boxed{M}
]

Hence,

[
\boxed{\text{Rahul receives internal marks.}}
]

Therefore, the goal is successfully proved using the Resolution Algorithm**.**

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books using the Resolution Algorithm.

1. Representation of the Knowledge Base in Propositional Logic

Let:

- (M) = Priya is a library member.
- (B) = Priya can borrow books.

Statement 1

If a person is a library member, then the person can borrow books.

$$
M \rightarrow B
$$

Statement 2

Priya is a library member.

$$
M
$$

Goal

$$
B
$$

2. Conversion of the Statements into CNF

The implication rule is:

$$
P \rightarrow Q \equiv \neg P \lor Q
$$

Therefore,

[
M \rightarrow B
]

becomes:

[
\neg M \lor B
]

The second statement is already in CNF:

[
M
]

Therefore, the CNF clauses are:

[
C_1 = \neg M \lor B
]

[
C_2 = M
]

3. Negation of the Goal

The goal is:

[
B
]

Negate the goal:

[
\neg B
]

Therefore, the complete set of clauses for resolution is:

$$
C_1 = \neg M \lor B
$$

$$
C_2 = M
$$

$$
C_3 = \neg B
$$

4. Resolution Algorithm

Resolution Step 1

Resolve C_1 and C_2:

$$
C_1 = \neg M \lor B
$$

$$
C_2 = M
$$

The complementary literals are:

$$
\neg M \quad \text{and} \quad M
$$

Applying the Resolution Rule:

$$
\frac{\neg M \lor B \qquad M}{B}
$$

Therefore, the new clause is:

[
\boxed{C_4 = B}
]

Resolution Step 2

Now resolve (C_4) with the negated goal (C_3):

[
C_4 = B
]

[
C_3 = \neg B
]

The complementary literals are:

[
B \quad \text{and} \quad \neg B
]

Applying the Resolution Rule:

[
\frac{B \qquad \neg B}{\boxed{\square}}
]

Therefore, we obtain the Empty Clause**:**

[
\boxed{\square}
]

5. Complete Resolution Proof

[
\frac{\neg M \lor B \qquad M}{B}
]

Then:

[
\frac{B \qquad \neg B}{\boxed{\square}}
]

Thus:

[
\boxed{\square}
]

The Empty Clause is obtained, so the negation of the goal produces a contradiction.

6. Final Conclusion

Since the Empty Clause (\square) is obtained, the original goal is proved.

[
\boxed{B}
]

Therefore,

[
\boxed{\text{Priya can borrow books.}}
]

Hence, the goal is successfully proved using the Resolution Algorithm.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement using the Resolution Algorithm.

1. Representation in Propositional Logic

Let:

- (A) = Arun cleared the aptitude test.
- (P) = Arun is eligible for placement.

Statement 1

If a student clears the aptitude test, then the student is eligible for placement.

[
A \rightarrow P
]

Statement 2

Arun cleared the aptitude test.

[
A
]

Goal

[
P
]

2. Conversion into CNF

The implication rule is:

$$
P \rightarrow Q \equiv \neg P \lor Q
$$

Therefore,

$$
A \rightarrow P
$$

becomes:

$$
\neg A \lor P
$$

The second statement is already in CNF:

$$
A
$$

Therefore, the CNF clauses are:

$$
\boxed{C_1 = \neg A \lor P}
$$

$$
\boxed{C_2 = A}
$$

3. Negation of the Goal

The goal is:

$$
P
$$

Negate the goal:

[
\neg P
]

Therefore, the complete set of clauses is:

[
C_1 = \neg A \lor P
]

[
C_2 = A
]

[
C_3 = \neg P
]

4. Resolution Algorithm – Step by Step

Resolution Step 1

Resolve (C_1) and (C_2):

[
C_1 = \neg A \lor P
]

[
C_2 = A
]

The complementary literals are:

[
\neg A \quad \text{and} \quad A
]

Applying the Resolution Rule:

[
\frac{\neg A \lor P \qquad A}{P}
]

Therefore, the new clause is:

[
\boxed{C_4 = P}
]

Resolution Step 2

Now resolve (C_4) with the negated goal (C_3):

[
C_4 = P
]

[
C_3 = \neg P
]

The complementary literals are:

[
P \quad \text{and} \quad \neg P
]

Applying the Resolution Rule:

[
\frac{P \qquad \neg P}{\boxed{\square}}
]

Therefore, we obtain the Empty Clause:

[
\boxed{\square}
]

5. Resolution Tree

The resolution tree can be represented as follows:

[
\boxed{\neg A \lor P}
\qquad
\boxed{A}
]

[
\hspace{45mm}\downarrow
]

[
\boxed{P}
]

Now use the negated goal:

[
\boxed{P}
\qquad
\boxed{\neg P}
]

[
\hspace{45mm}\downarrow
]

[
\boxed{\square}
]

Resolution Tree in Compact Form

[
\frac{
\frac{\neg A \lor P \qquad A}{P}
\qquad
\neg P

}{
\boxed{\square}
}
]

6. Final Conclusion

The resolution process produces the Empty Clause (\square)**.**

This means that the negation of the goal, (\neg P), contradicts the given knowledge base.

Therefore, the original goal is proved:

[
\boxed{P}
]

Hence,

[
\boxed{\text{Arun is eligible for placement.}}
]

Therefore, Arun is eligible for placement, and the goal is successfully proved using the Resolution Algorithm.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access using the Resolution Algorithm.

1. Representation of the Knowledge Base in Propositional Logic

Let:

- (P) = The user entered the correct password.
- (A) = The user is authenticated.
- (G) = The user is granted access.

Statement 1

If a user enters the correct password, then the user is authenticated.

[
P \rightarrow A
]

Statement 2

If a user is authenticated, then the user is granted access.

[
A \rightarrow G
]

Statement 3

The user entered the correct password.

[
P
]

Goal

$$
G
$$

2. Conversion of All Implications into CNF

The implication rule is:

$$
X \rightarrow Y \equiv \neg X \lor Y
$$

Therefore:

Statement 1

$$
P \rightarrow A
$$

becomes:

$$
\neg P \lor A
$$

Statement 2

$$
A \rightarrow G
$$

becomes:

$$
\neg A \lor G
$$

Statement 3

[
P
]

is already in CNF.

Therefore, the CNF clauses are:

[
\boxed{C_1 = \neg P \lor A}
]

[
\boxed{C_2 = \neg A \lor G}
]

[
\boxed{C_3 = P}
]

3. Negation of the Goal

The goal is:

[
G
]

Negate the goal:

[
\neg G
]

Therefore, the complete set of clauses for resolution is:

[
C_1 = \neg P \lor A
]

$$
C_2 = \neg A \lor G
$$

$$
C_3 = P
$$

$$
C_4 = \neg G
$$

4. Apply the Resolution Algorithm

Resolution Step 1

Resolve C_1 and C_3:

$$
C_1 = \neg P \lor A
$$

$$
C_3 = P
$$

The complementary literals are:

$$
\neg P \quad \text{and} \quad P
$$

Applying the Resolution Rule:

$$
\frac{\neg P \lor A \qquad P}{A}
$$

Therefore, the new clause is:

$$
\boxed{C_5 = A}
$$

This proves that the user is authenticated.

Resolution Step 2

Now resolve (C_2) and (C_5):

$$
C_2 = \neg A \lor G
$$

$$
C_5 = A
$$

The complementary literals are:

$$
\neg A \quad \text{and} \quad A
$$

Applying the Resolution Rule:

$$
\frac{\neg A \lor G \qquad A}{G}
$$

Therefore, the new clause is:

$$
\boxed{C_6 = G}
$$

This proves that the user is granted access.

Resolution Step 3

Now resolve (C_6) with the negated goal (C_4):

$$
C_6 = G
$$

$$
C_4 = \neg G
$$

The complementary literals are:

$$
G \quad \text{and} \quad \neg G
$$

Applying the Resolution Rule:

$$
\frac{G \qquad \neg G}{\boxed{\square}}
$$

Therefore, we obtain the Empty Clause**:**

$$
\boxed{\square}
$$

5. Complete Resolution Proof

The complete resolution process is:

First Resolution

$$
\frac{\neg P \lor A \qquad P}{A}
$$

Second Resolution

$$
\frac{\neg A \lor G \qquad A}{G}
$$

Third Resolution

[
\frac{G \qquad \neg G}{\boxed{\square}}
]

Thus:

[
\boxed{\square}
]

The Empty Clause is obtained.

6. Resolution Tree

The resolution tree can be represented as:

[
\boxed{\neg P \lor A}
\qquad
\boxed{P}
]

[
\hspace{40mm}\downarrow
]

[
\boxed{A}
]

Then:

[
\boxed{\neg A \lor G}
\qquad
\boxed{A}
]

[
\hspace{40mm}\downarrow
]

[
\boxed{G}
]

Finally, using the negated goal:

[
\boxed{G}
\qquad
\boxed{\neg G}
]

[
\hspace{40mm}\downarrow
]

[
\boxed{\square}
]

Compact Resolution Tree

[
\frac{
\frac{
\neg P \lor A \qquad P
}{A}
\qquad
\neg A \lor G
}{G}
]

Then:

[
\frac{G \qquad \neg G}{\boxed{\square}}
]

7. Final Conclusion

The Empty Clause (\square) is obtained after three resolution steps.

This means that the negation of the goal, (\neg G), leads to a contradiction with the knowledge base.

Therefore, the original goal is proved:

[
\boxed{G}
]

Hence,

[
\boxed{\text{The user is granted access.}}
]

Therefore, the user enters the correct password, becomes authenticated, and is consequently granted access. The goal is successfully proved using the Resolution Algorithm.